HIMANSHU YADAV

AI / ML Developer | LLM Agents | B.Tech Computer Science and Engineering
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SUMMARY

B.Tech Computer Science and Engineering student at IIIT Raichur (graduating 2027) building AI systems in Python, PyTorch, and TensorFlow. Built AURA, an LLM-powered research agent with retrieval, planner, researcher, and critic components, and GridPINN, a physics-informed neural network for AC optimal power flow. Project experience in deep learning, computer vision, federated learning, and edge deployment, with metric-driven model evaluation. Strong foundation in data structures, algorithms, and problem solving.

EDUCATION

IIIT Raichur – B.Tech Computer Science and Engineering CGPA: 7.1/10 Aug 2023 – May 2027

Relevant coursework: Data Structures and Algorithms, Machine Learning, Deep Learning, Computer Vision, DBMS, Operating Systems, OOP, Computer Networks

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript
LLM / Generative AI: LLM APIs, LLM Agents, Retrieval-Augmented Generation, Prompt Engineering, Embeddings, Semantic Search
AI / Machine Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, LightGBM, CNN, RNN, LSTM, Transfer Learning, Attention Mechanisms, Federated Learning, Differential Privacy
Computer Vision / NLP: OpenCV, scikit-image, YOLOv5, Object Detection, Image Classification, Segmentation, Super-Resolution, Text Classification, TF-IDF, Text Chunking
Model Optimization / Edge AI: INT8 Quantization, Pruning, Knowledge Distillation, TensorFlow Lite, ONNX
Optimization / Scientific Computing: Physics-Informed Neural Networks, CVXPY, pandapower, AC and DC Optimal Power Flow
Data: Pandas, NumPy, SciPy, Feature Engineering, PCA, Matplotlib, Seaborn
Model Evaluation: Accuracy, F1-Score, AUC-ROC, IoU, PSNR, SSIM, Cross-Validation
Backend / Tools: FastAPI, Git, Docker, Linux, Jupyter Notebook, TensorBoard, Google Colab

PROJECTS

AURA – LLM-Powered Research Agent Python, LLM APIs, FastAPI, Docker 2026

- Developed LLM-powered planner, researcher, and critic agents that retrieve evidence from documents and produce citation-backed research reports.
- Built document loading, text chunking, keyword search, and embedding-based semantic search for evidence retrieval; served the system through a FastAPI backend packaged with Docker.

GridPINN – Physics-Informed Neural Networks for AC Optimal Power Flow PyTorch, pandapower, CVXPY In progress

- Developing a physics-informed neural network in PyTorch to predict generator dispatch and bus voltages for AC optimal power flow, with power-balance and constraint terms in the training loss.
- Benchmarking the model against a standard neural-network baseline, DC-OPF (CVXPY), and exact AC-OPF solutions (pandapower), and measuring constraint violations.

Privacy-Preserving Video Surveillance – CNN & LSTM with Federated Learning TensorFlow Federated, Python 2025

- Built a distributed video analytics system combining CNN and LSTM under Federated Learning with Differential Privacy, reaching accuracy within 3% of the centralized baseline.
- Implemented Gaussian-noise differential privacy for on-device data processing and integrated real-time anomaly detection with alert generation.

Embedded AI Model Compression for LSTM Time-Series (TinyML) TensorFlow Lite, PyTorch, ONNX 2025

- Compressed an LSTM model 4x using INT8 quantization, structured pruning, and knowledge distillation, with under 2% accuracy loss; deployed on edge hardware at 30+ FPS.

Satellite Image Super-Resolution using Deep Learning TensorFlow, Keras, CNN, OpenCV 2025

- Implemented SRCNN and EDSR-inspired CNN architectures for 4x upscaling, achieving a 3–5 dB PSNR improvement and an SSIM of 0.92 on Sentinel-2 and Landsat imagery.

Privacy-Preserving Federated Learning for Healthcare TensorFlow Federated, Python, Pandas 2025

- Built a federated averaging framework with formal epsilon-delta differential privacy guarantees, within 2% of centralized model accuracy.

Drug Side Effect Prediction & Image Forgery Detection XGBoost, Scikit-Learn, TensorFlow, RDKit 2024–2025

- Drug prediction: 87% accuracy, 0.82 F1, 0.89 AUC-ROC on SIDER using XGBoost and neural networks with molecular fingerprint features.
- Forgery detection: 92% binary accuracy and 0.78 IoU segmentation using a multi-task CNN with Error Level Analysis on CASIA.

EXPERIENCE

Machine Learning Researcher *IIIT Raichur* 2023 – Present

- Researched deep learning, computer vision, federated learning, and TinyML, building 7+ AI projects in Python, TensorFlow, PyTorch, and Scikit-Learn.
- Applied INT8 quantization, structured pruning, and knowledge distillation to deploy models on NVIDIA Jetson Nano and MCU targets.

Teaching Assistant – Engineering Mathematics, DBMS, OS, Java OOP Labs *IIIT Raichur* 2024 – Present

- Designed and evaluated lab assignments for 60+ students covering databases, process scheduling, and object-oriented programming.

ACHIEVEMENTS

- **Reliance Foundation Scholar 2023** – Merit-based national scholarship for academic excellence in technology.
- **GIAN Summer Course Participant** – IIT Indore, Summer 2025: Ministry of Education program on Solving Linear Systems using Recurrent Neural Networks.